

ALY 6040 Data Mining Applications

Module 1 Final Project – Find Dataset Proposal

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1. Proposed Dataset Description

The dataset I have chosen from Kaggle.com, for my analysis comes from the nature of business that I have worked in for 4 years which is Employee Resource Management. This dataset on the Employee Performance and Productivity contains employee performance and productivity data with a total of **100,000 records** and **20 variables**. The dataset consists of both numerical and categorical variables, providing insights into various aspects of employee performance, satisfaction, and work behavior.

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a. Loading necessary libraries

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.metrics import classification_report, roc_auc_score
```

Purpose of this code:

- i. **Data Preprocessing:** Standardizes numerical data and encodes categorical data using transformers like StandardScaler and OneHotEncoder.
- ii. **Hyperparameter Tuning:** Uses GridSearchCV to find the best hyperparameters for the model building.
- iii. **Model Evaluation:** Evaluates model performance using metrics like classification_report and roc_auc_score.

2. Summary and Insights

a. Loading the Dataset

b. Getting Summary and general info of the dataset

```
Emp_Performamce = pd.read_csv('/content/Extended_Employee_Performance_and_Productivity_Data.csv')
summary = Emp_Performamce.describe()
info = Emp_Performamce.info()

print(summary)
print(info)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 20 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Employee_ID           100000 non-null int64
 1   Department            100000 non-null object
 2   Gender                100000 non-null object
```

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```

3   Age                100000 non-null int64
4   Job_Title          100000 non-null object
5   Hire_Date          100000 non-null object
6   Years_At_Company    100000 non-null int64
7   Education_Level     100000 non-null object
8   Performance_Score   100000 non-null int64
9   Monthly_Salary      100000 non-null float64
10  Work_Hours_Per_Week 100000 non-null int64
11  Projects_Handled     100000 non-null int64
12  Overtime_Hours      100000 non-null int64
13  Sick_Days           100000 non-null int64
14  Remote_Work_Frequency 100000 non-null int64
15  Team_Size           100000 non-null int64
16  Training_Hours      100000 non-null int64
17  Promotions          100000 non-null int64
18  Employee_Satisfaction_Score 100000 non-null float64
19  Resigned            100000 non-null bool

```

dtypes: bool(1), float64(2), int64(12), object(5)

memory usage: 14.6+ MB

	Employee_ID	Age	Years_At_Company	Performance_Score
\				
count	100000.000000	100000.000000	100000.000000	100000.000000
mean	50000.500000	41.029410	4.476070	2.995430
std	28867.657797	11.244121	2.869336	1.414726
min	1.000000	22.000000	0.000000	1.000000
25%	25000.750000	31.000000	2.000000	2.000000
50%	50000.500000	41.000000	4.000000	3.000000
75%	75000.250000	51.000000	7.000000	4.000000
max	100000.000000	60.000000	10.000000	5.000000

	Monthly_Salary	Work_Hours_Per_Week	Projects_Handled
Overtime_Hours \			
count	100000.000000	100000.000000	100000.000000
mean	6403.211000	44.956950	24.431170
std	1372.508717	8.942003	14.469584
min	3850.000000	30.000000	0.000000
25%	5250.000000	37.000000	12.000000
50%	6500.000000	45.000000	24.000000
75%	7500.000000	53.000000	37.000000
max	9000.000000	60.000000	49.000000

**The output has been reduced to avoid multiple documentation*

The variables and their data types are:

1. **Employee_ID** (int64): Unique identifier for each employee.
2. **Department** (object): The department the employee works in (e.g., Sales, Engineering).
3. **Gender** (object): The gender of the employee.
4. **Age** (int64): Age of the employee in years.
5. **Job_Title** (object): Job title held by the employee.
6. **Hire_Date** (object): Date the employee was hired.
7. **Years_At_Company** (int64): Number of years the employee has worked at the company.
8. **Education_Level** (object): Level of education attained by the employee (e.g., Bachelor's, Master's).
9. **Performance_Score** (int64): Performance rating (1 to 5) given to the employee.
10. **Monthly_Salary** (float64): Employee's monthly salary in USD.
11. **Work_Hours_Per_Week** (int64): Number of hours the employee works per week.
12. **Projects_Handled** (int64): The number of projects the employee has managed or participated in.
13. **Overtime_Hours** (int64): Number of overtime hours worked.
14. **Sick_Days** (int64): Number of sick days taken by the employee.
15. **Remote_Work_Frequency** (int64): Percentage of time the employee works remotely (0-100%).
16. **Team_Size** (int64): The number of team members under the employee's supervision or in their team.
17. **Training_Hours** (int64): Number of training hours completed by the employee.
18. **Promotions** (int64): Number of promotions the employee has received.
19. **Employee_Satisfaction_Score** (float64): Employee's satisfaction score (1 to 5), where 5 indicates highest satisfaction.
20. **Resigned** (bool): Indicates whether the employee has resigned (True/False).

c. Identifying Missing Values if Any

```
missing_values = Emp_Performance.isnull().sum()
print("Missing values per column:")
print(missing_values)
```

Missing values per column:

Employee_ID	0
Department	0
Gender	0
Age	0
Job_Title	0
Hire_Date	0
Years_At_Company	0
Education_Level	0
Performance_Score	0
Monthly_Salary	0
Work_Hours_Per_Week	0
Projects_Handled	0
Overtime_Hours	0
Sick_Days	0
Remote_Work_Frequency	0

```
Team_Size          0
Training_Hours     0
Promotions         0
Employee_Satisfaction_Score  0
Resigned           0
dtype: int64
```

The dataset has **no missing values** in any of the columns, as indicated by the zero counts for missing values across all variables.

Implications:

- i. **Data Integrity:** The absence of missing values suggests the dataset is complete, which simplifies the analysis since no imputation or removal of records is required.
- ii. **Direct Analysis:** We can proceed with analysis and modeling without concern for missing data, ensuring that the models will have full access to all data points.
- iii. **Next Steps:** Since there are no missing values, we can move directly into exploratory data analysis (EDA), such as identifying correlations, distributions of key variables, and applying machine learning models if needed.

3. Initial Interest in the Dataset

This dataset provides a comprehensive view of employee performance metrics, work behaviors, and job satisfaction, making it suitable for analyzing key factors that influence productivity, performance, and retention. I'm particularly interested in exploring how various factors such as work hours, team size, and overtime hours correlate with performance scores and employee satisfaction. Additionally, the "Resigned" column will be useful in predicting employee attrition and identifying potential risks of resignation based on performance, satisfaction, and work-related variables.

4. Goals of the Analysis:

- a. **Performance Analysis:** Identify the variables most strongly associated with high performance scores, such as education level, work hours, or projects handled.
- b. **Employee Retention:** Explore patterns that may predict employee resignation, such as overtime hours, satisfaction score, and promotion history.
- c. **Satisfaction Insights:** Understand how factors like remote work frequency, team size, and training hours impact employee satisfaction, and whether satisfied employees tend to stay longer with the company.

These insights could be valuable for human resources in optimizing employee retention strategies, improving work-life balance, and fostering an environment that promotes productivity and satisfaction.

5. References

- Data Source : Kaggle

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<https://www.kaggle.com/datasets/mexwell/employee-performance-and-productivity-data>

- Initial Analysis : Google Collab

<https://colab.research.google.com>